

JavaScript Validation API

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Constraint Validation DOM Methods

The Constraint Validation API provides methods for validating form elements:

Method	Description
<code>checkValidity()</code>	Returns true if an element contains valid data
<code>setCustomValidity()</code>	Sets a custom validation message

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Validation API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Constraint Validation DOM Methods</h4>

<input type="text" id="id1" required minlength="3" placeholder="Enter at least 3 chars">
<button onclick="validate()">Check Validity</button>
<p id="demo"></p>

<script>
function validate() {
    const input = document.getElementById("id1");

    if (input.checkValidity()) {
        document.getElementById("demo").innerHTML = "Input is valid!";
    } else {
        document.getElementById("demo").innerHTML = "Input is invalid. " + input.validationMessage;
    }
}
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

Constraint Validation DOM Properties

Form elements have these validation properties:

Property	Description
validity	Returns a ValidityState object
validationMessage	Returns the validation message
willValidate	Returns true if element will be validated

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Validation API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Constraint Validation DOM Properties</h4>

<input type="email" id="email" placeholder="Enter email address" required>
<button onclick="checkEmail()">Check Email</button>
<p id="demo2"></p>

<script>
function checkEmail() {
  const input = document.getElementById("email");
  let text = "Validation Properties:<br><br>";

  text += "willValidate: " + input.willValidate + "<br>";
  text += "validationMessage: " + input.validationMessage + "<br>";

  if (input.validity) {
    text += "validity.valid: " + input.validity.valid + "<br>";
  }

  if (input.value === "") {
    text += "<br>Please enter an email address.";
  }

  document.getElementById("demo2").innerHTML = text;
}
</script>

</body>
</html>

```

Example 2

Result [View Example](#)

Validity Properties

The `validity` object contains several boolean properties:

Property	Description
valueMissing	Element has no value (when required)
typeMismatch	Value doesn't match type (e.g., email)
patternMismatch	Value doesn't match pattern
tooLong	Value is too long
tooShort	Value is too short
rangeUnderflow	Value is below minimum
rangeOverflow	Value is above maximum
stepMismatch	Value doesn't match step
badInput	Value is invalid
customError	Custom validation error
valid	Element's value is valid

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Validation API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Validity Properties</h4>

<form id="myForm">
  <input type="text" id="username" required minlength="3" maxlength="10" placeholder="Username (
  <br><br>
  <input type="number" id="age" min="18" max="120" placeholder="Age (18-120)">
  <br><br>
  <input type="email" id="email2" placeholder="Email address">
  <br><br>
  <button type="button" onclick="showValidity()">Check All Fields</button>
</form>

<p id="demo3"></p>

<script>
function showValidity() {
  const fields = ["username", "age", "email2"];
  let text = "";

  for (let id of fields) {
    const input = document.getElementById(id);
    const v = input.validity;

    text += "<b>" + id + "</b>: ";
    if (v.valid) {
      text += "✓ Valid";
    } else {
      text += "✗ Invalid: ";
      if (v.valueMissing) text += "Required, ";
      if (v.typeMismatch) text += "Wrong type, ";
      if (v.patternMismatch) text += "No pattern match, ";
      if (v.tooShort) text += "Too short, ";
      if (v.tooLong) text += "Too long, ";
      if (v.rangeUnderflow) text += "Below minimum, ";
      if (v.rangeOverflow) text += "Above maximum, ";
      if (v.stepMismatch) text += "Wrong step, ";
      if (v.customError) text += "Custom error, ";
    }
    text += "<br>";
  }

  document.getElementById("demo3").innerHTML = text;
}

```

```
</script>
```

```
</body>
```

```
</html>
```

Example 3

Result [View Example](#)

The rangeOverflow Property

The `rangeOverflow` property is true when a value exceeds the `max` attribute:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Validation API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The rangeOverflow Property</h4>

<input type="number" id="quantity" max="100" placeholder="Max: 100">
<button onclick="checkRangeOverflow()">Check</button>
<p id="demo4"></p>

<script>
function checkRangeOverflow() {
  const input = document.getElementById("quantity");
  const value = input.value;

  if (value === "") {
    document.getElementById("demo4").innerHTML = "Please enter a value.";
    return;
  }

  if (input.validity.rangeOverflow) {
    document.getElementById("demo4").innerHTML =
      "Value " + value + " is too high! Max is " + input.max;
  } else {
    document.getElementById("demo4").innerHTML =
      "Value " + value + " is within range. Valid!";
  }
}
</script>

</body>
</html>
```

Example 4

Result [View Example](#)

The rangeUnderflow Property

The `rangeUnderflow` property is true when a value is less than the `min` attribute:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Validation API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The rangeUnderflow Property</h4>

<input type="number" id="score" min="50" placeholder="Min: 50">
<button onclick="checkRangeUnderflow()">Check</button>
<p id="demo5"></p>

<script>
function checkRangeUnderflow() {
  const input = document.getElementById("score");
  const value = input.value;

  if (value === "") {
    document.getElementById("demo5").innerHTML = "Please enter a value.";
    return;
  }

  if (input.validity.rangeUnderflow) {
    document.getElementById("demo5").innerHTML =
      "Value " + value + " is too low! Min is " + input.min;
  } else {
    document.getElementById("demo5").innerHTML =
      "Value " + value + " is within range. Valid!";
  }
}
</script>

</body>
</html>
```

Example 5

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Validation API](#)